

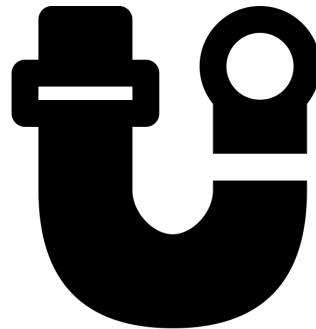
MAE 3127 - Fluid Mechanics Laboratory

Recitation-0

(P-Set01: Total points = 100)

_____, _____
Last Name First Name

Date



Created by Smashicons
from Noun Project

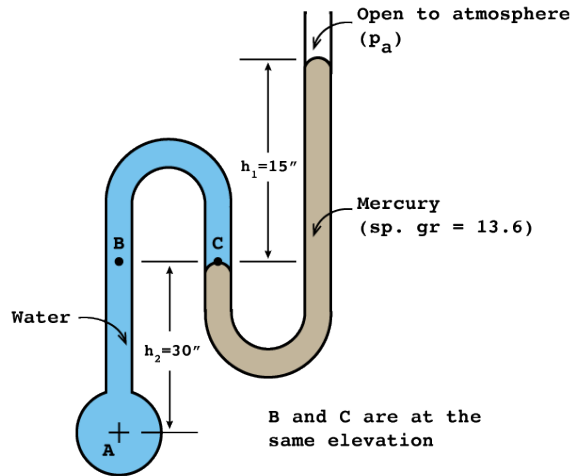
Figure 1: manometer by Smashicons from manometer Icons, Noun Project (CC BY 3.0)

According to the GWU Code of Academic Integrity, I pledge that I have neither given nor received unauthorized assistance on this work.

_____, _____
Student's Signature Date

P-Set01: Manometers

R0-P010: Determine the gage pressure [psig] at point A.



R0-P011: Two U-tube manometers measure the pressure in the same pressurized pipe. One manometer contains alcohol, and the other contains oil. The manometers are identical, other than the fluid they contain. The Alcohol manometer displays a standpipe vertical height of 5.80 cm, while the oil manometer displays a standpipe vertical height of 5.50 cm. Which of the following statements is correct:

- (a) Alcohol SG > oil SG.
- (b) Alcohol SG < oil SG.
- (c) SG for the alcohol does not vary as temperature changes.
- (d) Both the alcohol and oil SG are the same.

R0-P012: An inverted U-tube manometer containing oil is located between two reservoirs as shown. The left reservoir contains carbon tetrachloride (SG=1.59) and is closed and pressurized to 8 psig. The right reservoir is open to the atmosphere and contains freshwater with $\gamma = 62.4 \text{ lb}_f/\text{ft}^3$. Atmospheric pressure is standard. The distance H is 21.6 ft, determine the specific weight of the oil $[\text{lb}_f/\text{ft}^3]$. Note that H in the figure is not to scale.

